

Project Update Report

Overview

This report outlines my individual progress and technical contributions toward the Constraint Satisfaction Problem (CSP)-based Sudoku Solver project. My primary responsibility has been to design and develop the core CSP framework that represents the Sudoku problem. My work has focused on researching CSP concepts, creating the formal Sudoku model, developing the fundamental software components, and implementing the initial constraint validation system required for future solver development.

Week 1 (June 22 – June 28): CSP Research and Initial Design Development

- **CSP Theory Investigation:** I began by studying the fundamentals of Constraint Satisfaction Problems (CSPs) and how they are applied to solving Sudoku. The research focused on understanding the key elements of CSP, including variables, domains, and constraints, as well as the role of binary and n-ary constraints in representing relationships between different variables.
- **Initial System Planning:** After analysing the CSP approach, I prepared the initial design notes for the Sudoku solver. I identified the main components required for the system, including the representation of Sudoku cells as variables, possible values as domains, and Sudoku rules as constraints.
- **Architecture Preparation:** I outlined the planned structure of the CSP model and considered how the different components would interact with the solving algorithms that will be developed later.

Week 2 (June 29 – July 5): Sudoku CSP Modelling and Core Structure Design

- **Formal CSP Representation:** I formally modelled Sudoku as a Constraint Satisfaction Problem by defining all 81 cells of the Sudoku grid as variables. Each variable was assigned a domain containing possible values from 1 to 9.
- **Constraint Definition:** I designed the main constraint categories required for Sudoku validation:
 - **Row constraints** to ensure that each row contains unique values.
 - **Column constraints** to prevent repeated values within columns.
 - **Box constraints** to enforce the rules of the 3x3 subgrids.

- **Design Specification:** I documented the CSP design specification, including the relationship between variables, domains, constraints, and the overall grid structure. This specification will guide the implementation and integration stages of the project.

Week 3 (July 6 – July 15): Core CSP Implementation and Initial Validation

- **Core Class Development:** I started implementing the main classes required for the CSP framework:

- **Variable Class:** Developed to represent individual Sudoku cells and store their current state.
- **Domain Class:** Created to manage the possible values available for each variable.
- **Constraint Class:** Implemented to represent Sudoku rules and determine whether assignments satisfy the required conditions.
- **Grid Class:** Developed to represent the complete Sudoku board and manage the collection of variables.

- **Unit Testing:** I created initial unit tests for the implemented classes to verify that the core components were functioning correctly. The testing focused on checking object creation, value management, and interactions between CSP components.

- **Constraint Checking Development:** I began implementing the validation functions for checking row, column, and box constraints. Initial testing was performed using sample Sudoku configurations to ensure that invalid assignments could be identified.

- **Initial CSP Refactoring:** I started reviewing the CSP structure to prepare the model for future dynamic domain reduction and constraint propagation. This work is still in progress and requires further development before integration with the complete solver.